

Generating Nupzuki without its real images!

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1. Brief Description of the Visual Content

Various conceptual images of the character “Nupzuki”, generated by an image-to-image pipeline that utilized the SDXL base model output and multiple non-dedicated LoRAs. These LoRAs were intentionally used to create distinct stylistic variations, including “Nupzuki” in papercut and watercolor forms, showcasing design versatility without relying on a dedicated character model.



Figure 1. Nupzuki in various style

2. Technical Details

Our pipeline is as follows. First, we extract the core words and sentences that best represent “Nupzuki” and configure them as a text prompt. Using this prompt as a text condition, we perform conditional generation on the SDXL base model. Next, we select the images among the generated outputs that most accurately reflect the characteristics of “Nupzuki”. Finally, we apply a pre-trained LoRA which effectively expresses the desired style or background to the SDXL base model, using the selected images and the prompt(trigger words, etc.) as conditions to generate the final visual content. Figure 2 shows our pipeline.

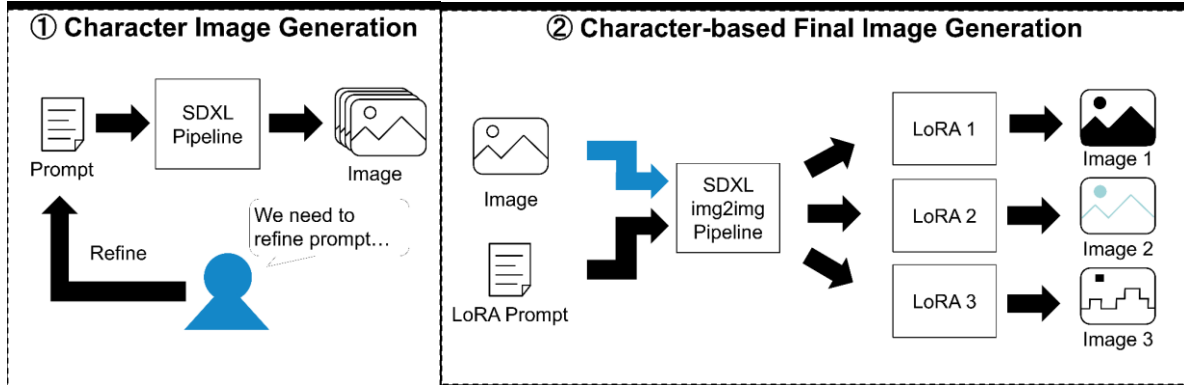


Figure 2. Pipeline of our work.

First point is **prompt engineering**. This stage is to establish a precise, high fidelity condition for the SDXL base model. The core engineering effort focused on structural specificity and constraint enforcement, utilizing explicit tokens such as “simple geometry”, “no mouth”, and “short height head” to compel the model toward a clearly defined images. The selection of structural features will be more discussed on Section 3. Our constraints such as “digital art” established the required high-fidelity rendering standard. Also, negative prompt such as “low quality”, “blurry”, and “distorted” which acts as a crucial technical safeguard for our visual generation.

Second point is **parameter tuning**. Our selection of number of steps considers the efficiency without losing its quality. CFG Scale which controls the guidance strength are also set to synthesize proper “Nupzuki” image without collapse. We also selected the images at the first step(text-to-image) to be the best representation of the character, and simultaneously the white normal background to make LoRA variations.

Last point is LoRA adaptation **controlling the LoRA_scale**. Every LoRA needed the different LoRA_scale for proper adaptation to “Nupzuki” images.

In this way, we made various “Nupzuki” image from the scratch **without referencing existing character images**.

3. Artistic and Visual Design Considerations

In designing the visual appearance of “Nupzuki”, we focused on identifying and preserving the essential traits that define the character. Reference image of “Nupzuki” is Figure 3. Our initial prompts were based on **direct visual observation**: core descriptors such as "long width head", short height head", "a tiny body", "two black eyes", and "no mouth." After evaluating early samples, we iteratively refined the prompt to include traits that were not fully captured by the previous prompt, such as the simplicity of the eyes.



Figure 3. Reference image of Nupzuki

Because mascot characters are generally perceived as “cute”, We intentionally incorporated the notion of cuteness as an aesthetic guidance. This subjective quality was translated into the prompt through descriptors that emphasize

friendliness, round shapes, and visual softness. The goal was to **align the generated character with the perceptual expectations** typically associated with mascots while preserving “Nupzuki”’s unique identity.

To maintain the fidelity of “Nupzuki”’s characteristics throughout the generation process, we minimized sources of visual distraction. All samples were generated without backgrounds so the model would focus on only the character form. Additionally, because too specific prompt constraints can cause deviations during synthesis, we used **relative descriptors** (e.g., "long", "short") simply and only when necessary.

We assumed that “Nupzuki” did not exist in the SDXL base model's training data. Therefore, we sampled outputs across a wide range of random seeds and manually selected those that most appropriately represented the intended character identity. Curated best sample was then used as image inputs for LoRA-based generation. During this stage, we explored multiple LoRA models and systematically tuned the LoRA scale to **balance "Nupzuki"'s core features with each LoRA's stylistic tendencies**. The final parameter choices were determined empirically based on **visual consistency, stylization strength, and preservation of character identity**.

After generating samples with eight pretrained LoRA models, we selected four outputs as the final results. The selection criteria included the degree to which **essential “Nupzuki” features were preserved**, the natural integration of each **LoRA's stylistic attributes**, and the **overall aesthetic quality** of the generated images. Final content is shown in **Figure 1**.

4. How to Reproduce Experiments

Unzip Team_22_code.zip or clone repository. with executing following commands in code directory, final result ‘team22_final_content.png’ will be generated in code directory.

Github link: <https://github.com/ShinJonghyeok/Generate-Nupzuki-without-its-real-images.git>

```
>> conda env create -f environment.yml
>> conda activate sdxl-team22
>> python 1_generate_nupzuki.py
>> python 2_generate_content.py
>> python 3_postprocess.py
```

5. References

- [1] Nupzuki image from email signature, <https://giving.kaist.ac.kr/nubzuki/KAIST-hi.gif>
- [2] huggingface Diffusers Stable Diffusion XL, https://huggingface.co/docs/diffusers/en/api/pipelines/stable_diffusion/stable_diffusion_xl
- [3] huggingface SDXL LoRA - PpromptEngineer/pe-ballon-diffusion-style, <https://huggingface.co/PromptEngineer/pe-balloon-diffusion-style>
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- [7] huggingface SDXL LoRA - jbilcke-hf/sdxl-zelda64 <https://huggingface.co/jbilcke-hf/sdxl-zelda64>
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- [11] SDXL,

Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., and Rombach, R. (2024). SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis.